# <u>Assignment NO 7:</u>

## Stored Procedures and Functions in PL/SQL.

## Write PLSQL Procedures and Function for given problem statements.

**<u>Name</u> - Harshada Suryawanshi**

**<u>Roll No</u> - 57**

**<u>PRN</u> - 1272250822**

**<u>Class</u> - SYCSE**

**<u>Panel</u> – C [ Batch C3 ]**

## Problem Statement:

**Q.1) Create a procedure named GenerateMonthlySalesSummary that doesn't take any input parameters. This procedure should automate the process of summarizing monthly sales data and storing it in a separate summary table for reporting purposes.**

**Here are the requirements:**

1. **The procedure should check if a MonthlySalesSummary table exists. If it doesn't, it should be created with columns for month_year (string), total_sales (decimal), and number_of_orders (integer).**
2. **It should then calculate the total sales and the total number of orders for the previous month from an Orders table, which includes order_date and total_amount columns.**
3. **The procedure must check if a summary record for the previous month already exists in the MonthlySalesSummarytable.**
4. **If the record already exists, it should be updated with the new calculated values.**
5. **If the record does not exist, a new record should be inserted.**
6. **The procedure should handle dates dynamically, always calculating for the month immediately preceding the current date.**

**INPUT & OUTPUT:**

```
mysql> create database db_sales;
Query OK, 1 row affected (0.01 sec)

mysql> use db_sales;
Database changed
mysql> create table if not exists orders (
    ->     order_id int primary key,
    ->     order_date date,
    ->     total_amount decimal(10,2)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> insert into orders (order_date, total_amount) values
    -> ('2026-02-05', 500.00),
    -> ('2026-02-10', 1200.00),
    -> ('2026-02-15', 800.00),
    -> ('2026-03-01', 700.00);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> delimiter $$
mysql> create procedure generatemonthlysalessummary()
    -> begin
    ->     declare prev_month_start date;
    ->     declare prev_month_end date;
    ->     declare v_total_sales decimal(10,2);
```

```
    ->     declare v_total_orders int;
    ->     declare month_year_str varchar(20);
    ->
    ->     set prev_month_start = date_format(date_sub(curdate(), interval 1 
month), '%Y-%m-01');
    ->     set prev_month_end = last_day(date_sub(curdate(), interval 1 month));
    ->
    ->     set month_year_str = date_format(prev_month_start, '%Y-%m');
    ->
    ->     create table if not exists monthlysalessummary (
    ->         month_year varchar(20) primary key,
    ->         total_sales decimal(10,2),
    ->         number_of_orders int
    ->     );
    ->
    ->     select
    ->         ifnull(sum(total_amount),0),
    ->         count(*)
    ->     into v_total_sales, v_total_orders
    ->     from orders
    ->     where order_date between prev_month_start and prev_month_end;
    ->
    ->     if exists (
    ->         select 1 from monthlysalessummary where month_year = 
month_year_str
    ->     ) then
    ->         update monthlysalessummary
```

```
    ->        set total_sales = v_total_sales,
    ->            number_of_orders = v_total_orders
    ->        where month_year = month_year_str;
    ->    else
    ->        insert into monthlysalessummary(month_year, total_sales,
number_of_orders)
    ->        values (month_year_str, v_total_sales, v_total_orders);
    ->    end if;
    ->
    -> end$$
Query OK, 0 rows affected (0.01 sec)
mysql> delimiter ;
mysql> call generatemonthlysalessummary();
Query OK, 1 row affected (0.03 sec)

mysql> select * from monthlysalessummary;
+------------+-------------+------------------+
| month_year | total_sales | number_of_orders |
+------------+-------------+------------------+
| 2026-02    |     2500.00 |                3 |
+------------+-------------+------------------+
1 row in set (0.00 sec)
```

**Q.2) Design a function named CheckProductAvailability that takes two input parameters: product_id (integer) and quantity_requested (integer). This function should return a boolean value (1 for true, 0 for false) indicating whether the requested quantity of a product is available in stock.**

**Here are the requirements:**

1. **The function should query a Products table, which contains product_id and stock_quantity columns.**
2. **It should check the stock_quantity for the given product_id.**
3. **The function should return 1 (true) if the stock_quantity is greater than or equal to the quantity_requested.**
4. **The function should return 0 (false) if the stock_quantity is less than the quantity_requested or if the product_id does not exist in the table.**

**INPUT & OUTPUT:**

```
mysql> create table if not exists products (
    ->     product_id int primary key,
    ->     stock_quantity int
    -> );
Query OK, 0 rows affected (0.04 sec)


mysql> insert into products values
    -> (1, 50),
    -> (2, 20),
    -> (3, 0);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql> delimiter $$
mysql> create function check_product_availability(p_product_id int,
quantity_requested int)
    -> returns int
    -> deterministic
    -> begin
    ->     declare result int;
    ->
    ->     select
    ->         case
    ->             when stock_quantity >= quantity_requested then 1
    ->             else 0
    ->         end
    ->     into result
    ->     from products
    ->     where product_id = p_product_id;
    ->
    ->     if result is null then
    ->         return 0;
    ->     end if;
    ->
    ->     return result;
    ->
    -> end$$
Query OK, 0 rows affected (0.01 sec)
mysql> delimiter ;
```

```
mysql> select check_product_availability(1, 10);
+-----------------------------------+
| check_product_availability(1, 10) |
+-----------------------------------+
|                                 1 |
+-----------------------------------+
1 row in set (0.01 sec)

mysql> select check_product_availability(2, 30);
+-----------------------------------+
| check_product_availability(2, 30) |
+-----------------------------------+
|                                 0 |
+-----------------------------------+
1 row in set (0.00 sec)

mysql> select check_product_availability(3, 1);
+----------------------------------+
| check_product_availability(3, 1) |
+----------------------------------+
|                                0 |
+----------------------------------+
1 row in set (0.00 sec)

mysql> select check_product_availability(5, 10);
+-----------------------------------+
```

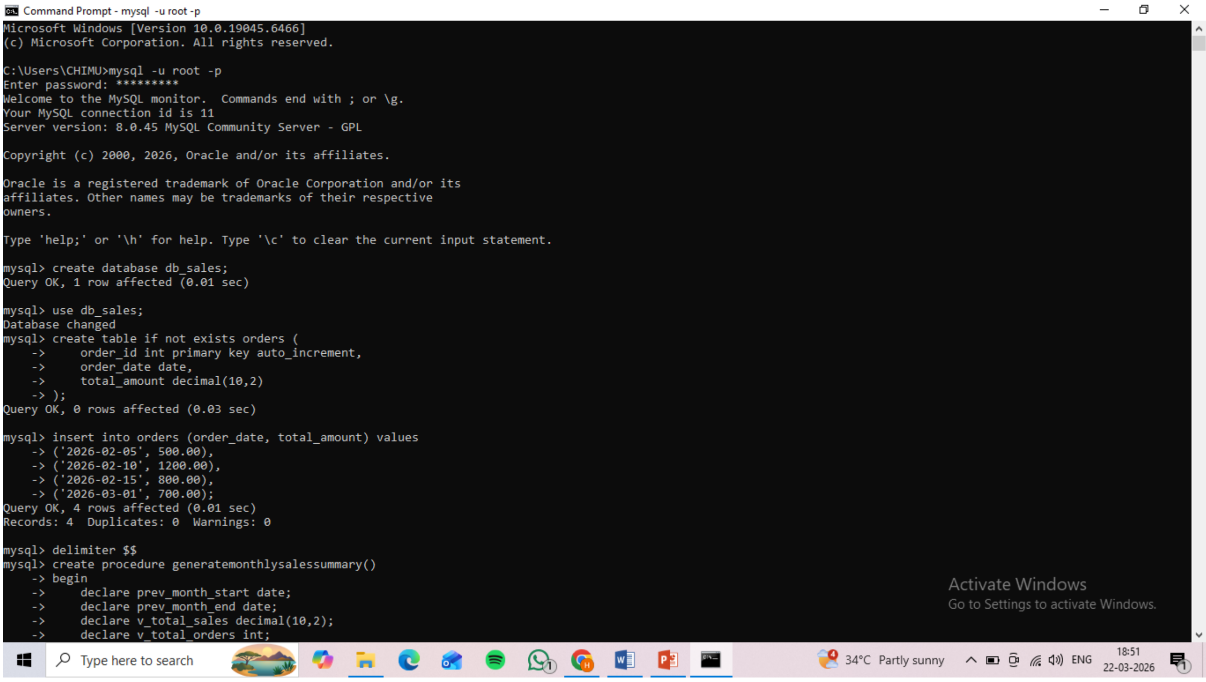
| check_product_availability(5, 10) |

+----------------------------------+

| 0 |

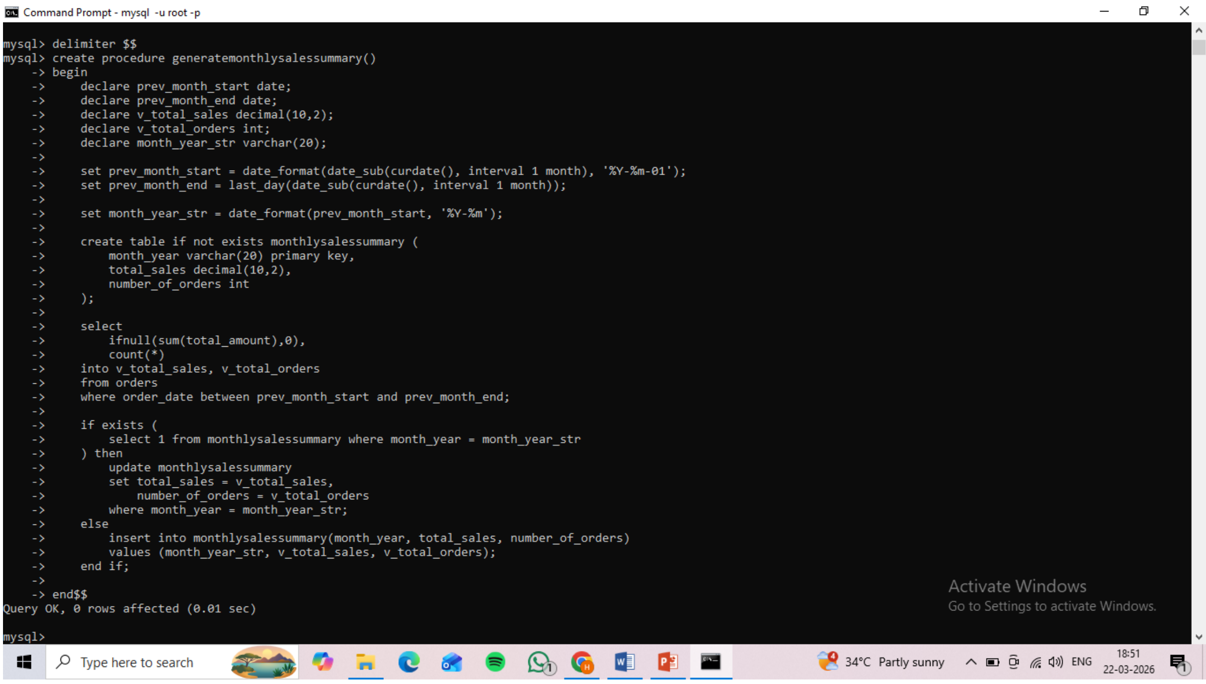
+----------------------------------+

1 row in set (0.00 sec)



```
Command Prompt - mysql -u root -p
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CHIMU>mysql -u root -p
Enter password: *********
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 11
Server version: 8.0.45 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database db_sales;
Query OK, 1 row affected (0.01 sec)

mysql> use db_sales;
Database changed
mysql> create table if not exists orders (
    ->     order_id int primary key auto_increment,
    ->     order_date date,
    ->     total_amount decimal(10,2)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> insert into orders (order_date, total_amount) values
    -> ('2026-02-05', 500.00),
    -> ('2026-02-10', 1200.00),
    -> ('2026-02-15', 800.00),
    -> ('2026-03-01', 700.00);
Query OK, 4 rows affected (0.01 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> delimiter $$
mysql> create procedure generatemonthlysalessummary()
    -> begin
    ->     declare prev_month_start date;
    ->     declare prev_month_end date;
    ->     declare v_total_sales decimal(10,2);
    ->     declare v_total_orders int;
```





```
Command Prompt - mysql -u root -p
mysql> delimiter $$
mysql> create procedure generatemonthlysalessummary()
    -> begin
    ->     declare prev_month_start date;
    ->     declare prev_month_end date;
    ->     declare v_total_sales decimal(10,2);
    ->     declare v_total_orders int;
    ->     declare month_year_str varchar(20);
    ->
    ->     set prev_month_start = date_format(date_sub(curdate(), interval 1 month), '%Y-%m-01');
    ->     set prev_month_end = last_day(date_sub(curdate(), interval 1 month));
    ->
    ->     set month_year_str = date_format(prev_month_start, '%Y-%m');
    ->
    ->     create table if not exists monthlysalessummary (
    ->         month_year varchar(20) primary key,
    ->         total_sales decimal(10,2),
    ->         number_of_orders int
    ->     );
    ->
    ->     select
    ->         ifnull(sum(total_amount),0),
    ->         count(*)
    ->     into v_total_sales, v_total_orders
    ->     from orders
    ->     where order_date between prev_month_start and prev_month_end;
    ->
    ->     if exists (
    ->         select 1 from monthlysalessummary where month_year = month_year_str
    ->     ) then
    ->         update monthlysalessummary
    ->         set total_sales = v_total_sales,
    ->             number_of_orders = v_total_orders
    ->         where month_year = month_year_str;
    ->     else
    ->         insert into monthlysalessummary(month_year, total_sales, number_of_orders)
    ->         values (month_year_str, v_total_sales, v_total_orders);
    ->     end if;
    ->
    -> end$$
Query OK, 0 rows affected (0.01 sec)

mysql>
```

Command Prompt - mysql -u root -p

```
    ->     select
    ->         ifnull(sum(total_amount),0),
    ->         count(*)
    ->     into v_total_sales, v_total_orders
    ->     from orders
    ->     where order_date between prev_month_start and prev_month_end;
    ->
    ->     if exists (
    ->         select 1 from monthlysalessummary where month_year = month_year_str
    ->     ) then
    ->         update monthlysalessummary
    ->         set total_sales = v_total_sales,
    ->             number_of_orders = v_total_orders
    ->         where month_year = month_year_str;
    ->     else
    ->         insert into monthlysalessummary(month_year, total_sales, number_of_orders)
    ->         values (month_year_str, v_total_sales, v_total_orders);
    ->     end if;
    ->
    -> end$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> delimiter ;
mysql> call generatemonthlysalessummary();
Query OK, 1 row affected (0.03 sec)

mysql> select * from monthlysalessummary;
+------------+-------------+------------------+
| month_year | total_sales | number_of_orders |
+------------+-------------+------------------+
| 2026-02    |     2500.00 |                3 |
+------------+-------------+------------------+
1 row in set (0.00 sec)

mysql>
```



Type here to search 34°C Partly sunny ENG 18:51 22-03-2026

Command Prompt - mysql -u root -p

```
mysql> create table if not exists products (
    ->     product_id int primary key,
    ->     stock_quantity int
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> insert into products values
    -> (1, 50),
    -> (2, 20),
    -> (3, 0);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> delimiter $$
mysql> create function checkproductavailability(p_product_id int, quantity_requested int)
    -> returns int
    -> deterministic
    -> begin
    ->     declare stock int;
    ->
    ->     select stock_quantity
    ->     into stock
    ->     from products
    ->     where product_id = p_product_id;
    ->
    ->     if stock is null then
    ->         return 0;
    ->     elseif stock >= quantity_requested then
    ->         return 1;
    ->     else
    ->         return 0;
    ->     end if;
    ->
    -> end$$
Query OK, 0 rows affected (0.01 sec)

mysql> delimiter ;
mysql> select checkproductavailability(1, 10);
+---------------------------------+
| checkproductavailability(1, 10) |
+---------------------------------+
|                               1 |
+---------------------------------+
```



Type here to search 34°C Partly sunny ENG 18:58 22-03-2026

Command Prompt - mysql -u root -p

```
    ->     else
    ->         return 0;
    ->     end if;
    ->
    -> end$$
Query OK, 0 rows affected (0.01 sec)

mysql> delimiter ;
mysql> select checkproductavailability(1, 10);
+---------------------------------+
| checkproductavailability(1, 10) |
+---------------------------------+
|                               1 |
+---------------------------------+
1 row in set (0.01 sec)

mysql> select checkproductavailability(2, 30);
+---------------------------------+
| checkproductavailability(2, 30) |
+---------------------------------+
|                               0 |
+---------------------------------+
1 row in set (0.00 sec)

mysql> select checkproductavailability(3, 1);
+--------------------------------+
| checkproductavailability(3, 1) |
+--------------------------------+
|                              0 |
+--------------------------------+
1 row in set (0.00 sec)

mysql> select checkproductavailability(5, 10);
+---------------------------------+
| checkproductavailability(5, 10) |
+---------------------------------+
|                               0 |
+---------------------------------+
1 row in set (0.00 sec)

mysql>
```

Command Prompt - mysql -u root -p

```
mysql> delimiter ;
mysql> delimiter $$
mysql>
mysql> create function check_product_availability(p_product_id int, quantity_requested int)
    -> returns int
    -> deterministic
    -> begin
    ->     declare result int;
    ->
    ->     select
    ->         case
    ->             when stock_quantity >= quantity_requested then 1
    ->             else 0
    ->         end
    ->     into result
    ->     from products
    ->     where product_id = p_product_id;
    ->
    ->     if result is null then
    ->         return 0;
    ->     end if;
    ->
    ->     return result;
    ->
    -> end$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> delimiter ;
mysql> select check_product_availability(1,10);
+----------------------------------+
| check_product_availability(1,10) |
+----------------------------------+
|                                1 |
+----------------------------------+
1 row in set (0.01 sec)

mysql> select check_product_availability(2,30);
+----------------------------------+
| check_product_availability(2,30) |
+----------------------------------+
|                                0 |
+----------------------------------+
1 row in set (0.00 sec)
```

Activate Windows
Go to Settings to activate Windows.

Type here to search 34°C Mostly clear ENG 19:40 23-03-2026

Command Prompt - mysql -u root -p

```
    -> end$$
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> delimiter ;
mysql> select check_product_availability(1,10);
+----------------------------------+
| check_product_availability(1,10) |
+----------------------------------+
|                                1 |
+----------------------------------+
1 row in set (0.01 sec)

mysql> select check_product_availability(2,30);
+----------------------------------+
| check_product_availability(2,30) |
+----------------------------------+
|                                0 |
+----------------------------------+
1 row in set (0.00 sec)

mysql> select check_product_availability(3,1);
+---------------------------------+
| check_product_availability(3,1) |
+---------------------------------+
|                               0 |
+---------------------------------+
1 row in set (0.00 sec)

mysql> select check_product_availability(5,10);
+----------------------------------+
| check_product_availability(5,10) |
+----------------------------------+
|                                0 |
+----------------------------------+
1 row in set (0.00 sec)

mysql> _
```

Activate Windows
Go to Settings to activate Windows.

Type here to search 34°C Mostly clear ENG 19:40 23-03-2026